# Base station physics: summary

Params: base/spec/params.py v1.3.2

### Spring (each, A228, SF 1.2 on Shigley static limits)

| | |
|---|---|
| Rate used / by formula | 0.125 / 0.117 N/mm |
| Initial tension used | 7.0 N |
| Body coils, mass | 98, 53 g |
| Safe force (governed by) | 40.9 N (hook torsion) |
| Safe extension (wire) / spec cap | 271 / 265 mm |
| Surge (fixed-fixed) | 24 Hz |

### Launch (chain spring model)

| | |
|---|---|
| Stroke, ratio, pitch | 241 mm, 1.0:1, 35 deg |
| Spring extension full / rest | 265 / 24 mm |
| Stored energy (2 springs) | 12.1 J |
| Carriage force at full draw | 80 N |
| Muzzle speed | 6.85 m/s |
| Peak acceleration, stroke time | 18 g, 59 ms |
| Efficiency (dart KE / stored) | 54 % |
| Carriage KE at stop, pad peak | 1.89 J, 285 N (19 mm pad) |
| Spring KE at stop, coil clash | 0.82 J, 8.0 m/s |

### Flight (bead-chain tether + reel model, level ground)

| | |
|---|---|
| Line path at the shot | taut (path only lengthens) |
| Range at 0.5 / 0.75 / 1.0 N brake | 4.18 / 3.76 / 3.39 m (pitch 34) |
| Same shot, no tether | 5.01 m |
| Flight time, apex (0.75 N) | 0.80 s, 1.11 m |
| Impact speed, angle (0.75 N) | 5.7 m/s, 55 deg down |
| Ballistics table | 6 to 47 deg, best 33.5 deg: 3.76 m |

### Storage reel payout (LIGHT 0.5 N at full reel)

| | |
|---|---|
| Reel empty mass, inertia full | 46 g, 31.4 kg mm2 |
| Line radius full / core | 30.7 / 25.0 mm (brake +23 % at core) |
| Line pull in the stroke (spin-up) | 5.05 N |
| Reel rpm max | 2129 rpm |
| Overrun after landing | 0.56 m in 0.27 s |

### Capstan and tow

| | |
|---|---|
| Wraps for tow limit at the capstan (65 N, mu 0.15) | 2.7 raw, 5 design |
| Take-up needed for the tow at the actual wraps (7) | 0.09 N |
| Tow at limit: torque, speed, current | 3.64 N m, 0.137 m/s, 5.50 A |
| Wiper at limiter (5 N m), 12.6 V: torque, current, line pull | 5.00 N m, 7.0 A, 71 N (break 667 N) |
| Wiper motor stall (no limiter), 12.6 V: torque, current, line pull | 14.1 N m, 17.0 A, 201 N |
| Tow sensing at the dart: wiper=13.1 V, tip above free | 2.3 N (capstan side x1.3) |

### Drivetrain (wiper at 11.1 V with sag)

| | |
|---|---|
| aim: peak torque, current, time | 0.33 N m, 1.9 A, 7 s |
| cock: peak torque, current, time | 2.93 N m, 4.7 A, 3 s |
| reel: peak torque, current, time | 0.43 N m, 2.0 A, 107 s |
| tow: peak torque, current, time | 3.64 N m, 5.5 A, 37 s |

### To measure (then edit base/spec/params.py and run ./run_all.sh)

1. Spring hang test: hang 1, 2, 3 kg from one spring and measure the length each time (hooks included). python calib.py spring FREE_MM 1:MM 2:MM 3:MM, then SPRING k_n_per_mm and F0_n. Every range number scales with the stored energy.
2. Wiper: free speed (turns in 30 s at 12.0 V), free current, stall torque (arm + luggage scale, 1 s) and stall current. python calib.py wiper --v 12 --rpm .. --i-free .. --stall-kg .. --arm-mm .. --i-stall .., then WIPER n0_rpm, stall_nm, stall_a, free_a.
3. Dart mass on a kitchen scale with ballast and tether knot (spec 0.279 kg), then DART mass_kg.
4. Slip clutch: luggage scale on the line with the reel full: LIGHT 0.5 N, FIRM 5 N (REEL clutch).
5. First shots on level ground: python calib.py range pitch:brake:range ..., then LAUNCH v_scale = FACTOR.